

1 · FORCING (Eulerian)

gridded fields read by OpenDrift readers

Ocean currents u, v (w)

GLORYS12 1/12° · CROCO 1/36°
CMEMS forecast

10 m wind U_{10}

ERA5 · GFS · ECMWF · WRF

Waves & Stokes drift

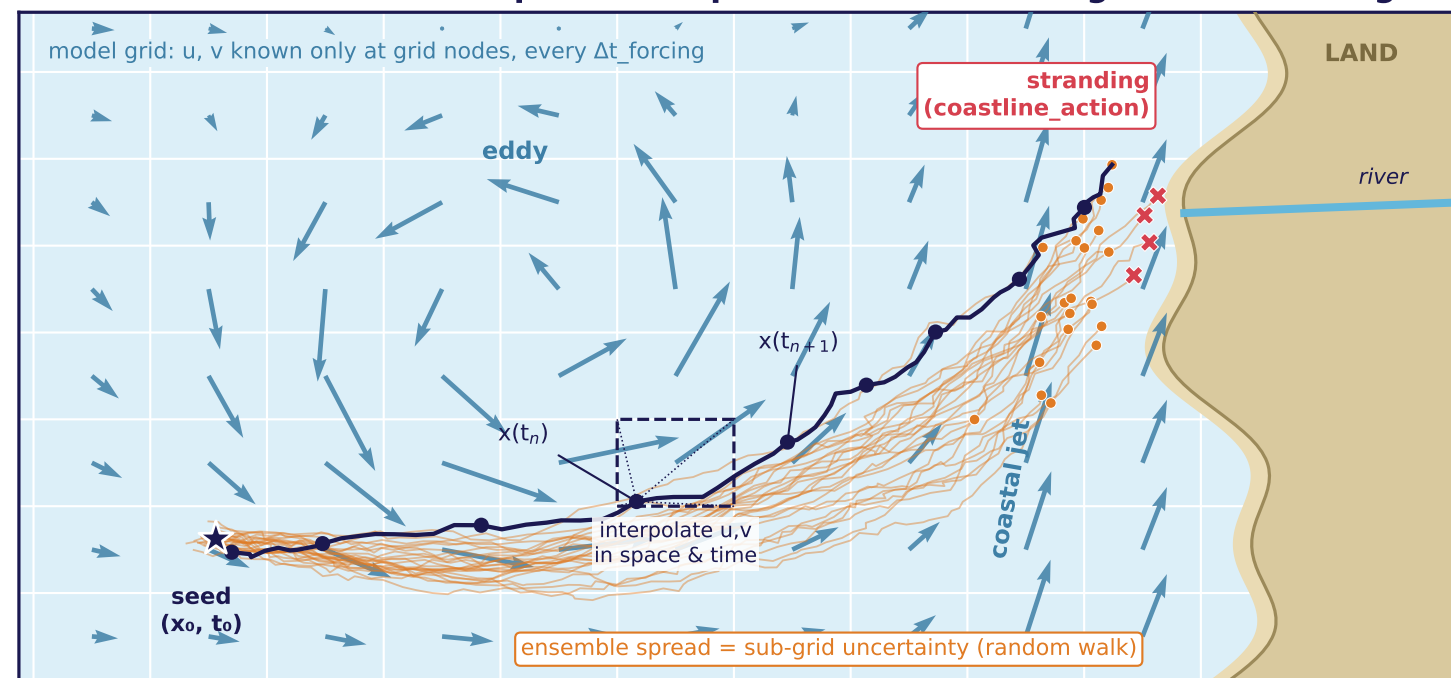
H_s , T_p , u_{Stokes}
(or estimated from wind)

Mixing & water column

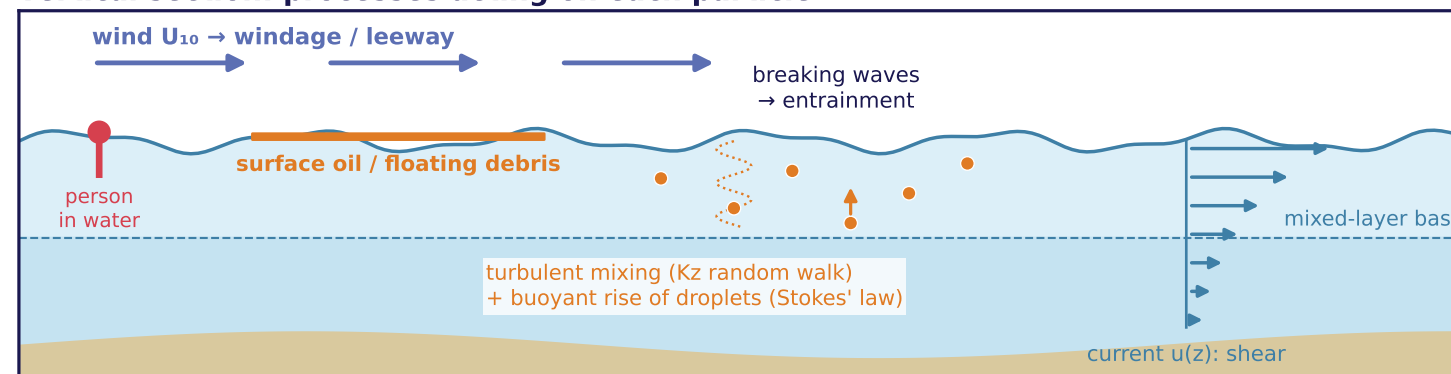
K_z , mixed layer, T , S ,
bathymetry, coastline

hindcast (reanalysis) or forecast

2 · LAGRANGIAN ENGINE — plan view: particles move through the Eulerian grid



vertical section: processes acting on each particle



$$\frac{dx}{dt} = \mathbf{u}_{\text{current}}(\mathbf{x}, t) + \alpha \mathbf{U}_{10} + \mathbf{u}_{\text{Stokes}} + \mathbf{u}_{\text{leeway}} \quad \mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{v} \Delta t + \sqrt{2K\Delta t} \boldsymbol{\xi}$$

advection (Euler / Runge-Kutta 4) + stochastic random walk for unresolved motions · $\alpha \approx 2-3\%$ windage · $\boldsymbol{\xi} \sim N(0, 1)$

3 · APPLICATIONS & PRODUCTS

same engine, different element types

Oil spill — OpenOil

evaporation · emulsification ·
natural dispersion · 2D vs 3D
 $\rightarrow$ oil budget, beached mass

Search & Rescue — Leeway

object-specific leeway coefficients
(person, raft, fishing vessel)
 $\rightarrow$ search area at $T + 24$ h

Marine plastics — OceanDrift

river sources (Sanaga, Nyong, Wouri)
windage · stranding / resuspension
 $\rightarrow$ coastal accumulation sectors

Diagnostics

time to first coastal impact · stranded fraction
convex-hull area · centroid separation
probability maps from ensembles

OpenDrift workflow

